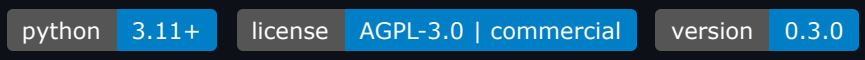
checocadena / oneapi-agent-toolkit

oneapi-agent-toolkit

Evidence-first tools that help coding agents finish CUDA → SYCL ports, and prove them.



Done only when verify passes and scan_helpers finds zero dpct helpers, on the same freshly compiled source.



Open-source software
Press kit

Checo Cadena
2026

What it is

Intel's SYCLomatic gets CUDA code most of the way to SYCL, an open standard not tied to NVIDIA. oneapi-agent-toolkit gives a coding agent the tools to finish that port and prove it: a port counts as done only when its output matches the original and no SYCLomatic helpers remain.

The toolkit is a Python package with five tools: `migrate`, `compile`, `scan_helpers`, `verify` and `benchmark`. Each returns evidence rather than a claim: a status, the exact commands that ran, parsed diagnostics, numbers and a list of limitations. Agents reach the tools through an MCP server, a command line that prints the same results, or a bundled agent loop that enforces the "done" rule itself. A GPU command rents a machine on one of four clouds, checks the port against the original CUDA built there, and deletes the machine.

The problem it answers was observed in the repository's own examples. On HeCBench's softmax, SYCLomatic produced code that compiled with zero warnings but had silently swapped `expf` for a reduced-precision function: 22,156 of 524,288 outputs were wrong on an AVX-512 CPU. Only running the code and comparing outputs caught it. Coding agents asked to finish such ports tend to stop as soon as the code compiles.

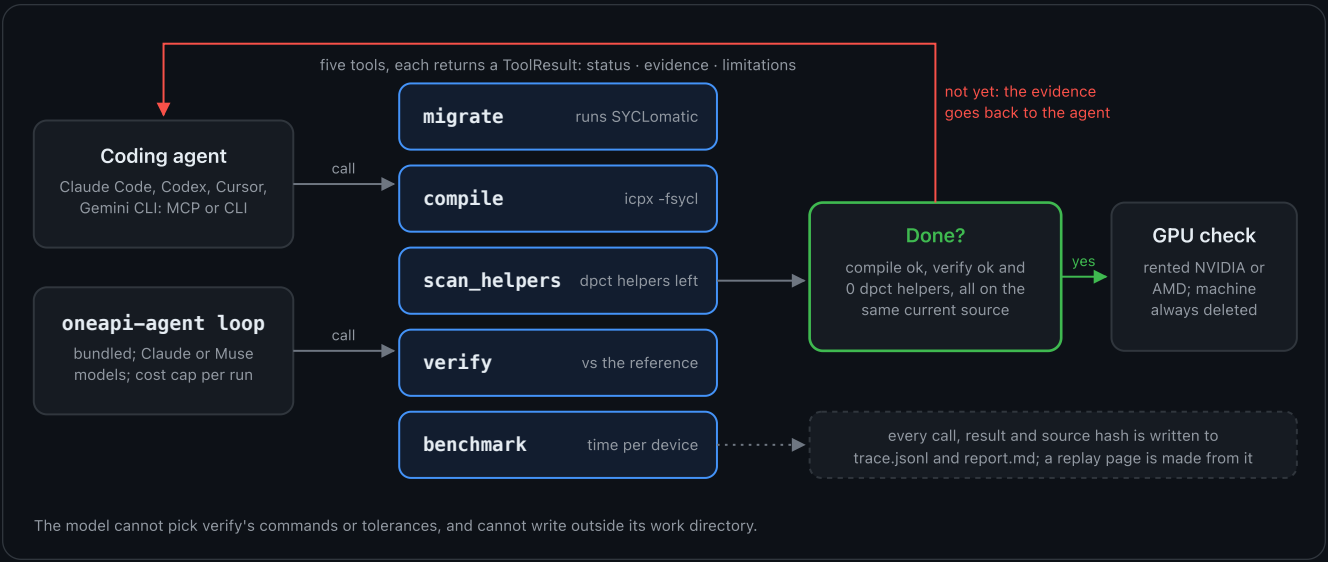
In 120 measured runs with two Claude models, agents using the toolkit finished every port correctly and with no helpers left (40 of 40). The same agents without its checks managed 23 of 40, and 14 of 40 when given only a shell.

Stage: version 0.3.0, a working research tool, not yet on PyPI. The first commit is dated 15 October 2026. All measured ports were made by the bundled loop with Claude models and graded on a CPU, on small single-file programs: indicative, not a benchmark. Intel GPUs have not been tested.

Used in Muse Port. Muse Port, by the same author, installs this toolkit as a pinned dependency for its tools, agent loop and GPU tests. Muse Port was built during the Meta Global AI Developer Hackathon 2026 and submitted there. The toolkit's state before the event is the git tag `pre_hackathon`; the 11 changes made to it for Muse Port are listed in Muse Port's repository.

TO CONFIRM (Checo): the repository shows no outside users, adoption or press. Add any that exist, or leave this kit as is.

How it works



`migrate` runs SYCLomatic and parses every warning, with the file and line in both the CUDA and the migrated code.

`compile` builds with Intel's `icpx -fsycl` and returns each diagnostic as data.

`scan_helpers` counts the `dpct::` helpers SYCLomatic leaves behind, by kind and file, with the plain SYCL to replace each, and flags risky patterns such as reduced-precision math.

`verify` runs the reference and the port on the same inputs and compares the outputs: error statistics for arrays, or a null test in dBFS for audio.

`benchmark` times a program on each SYCL device it finds and reports the raw samples.

`Honest failures.` If a compiler is missing, the result says `toolchain_missing` and names the command that could not run. Nothing is faked.

`The loop.` `oneapi-agent loop` lets a model call the tools until compile, verify and a zero-helper scan all hold on the same source. The model cannot choose verify's commands or tolerances, and each run has a cost cap.

`GPU check.` `remote-test` rents a GPU on Runpod, Vast.ai, Vultr or DigitalOcean, builds the original CUDA there, verifies the port against it and deletes the machine.

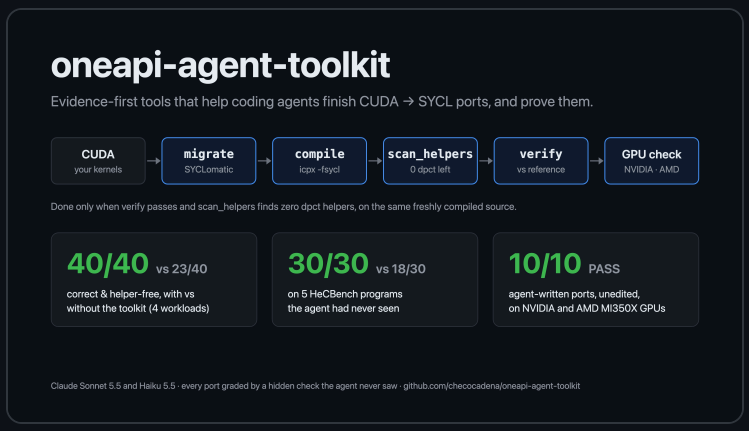
Results so far

From the repository's README and docs folder, as of 11 November 2026. Every number links there to the committed traces and sources.

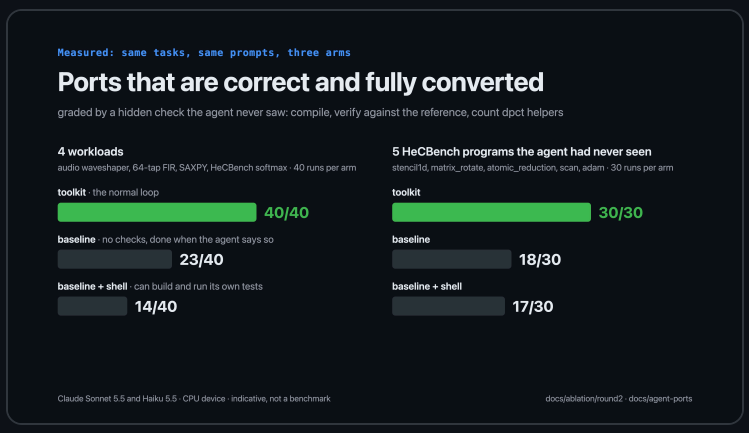
With vs without	4 workloads, Claude Sonnet and Haiku, 5 runs each, 120 runs in all. Correct and helper-free: toolkit **40/40**, no checks 23/40, a shell only 14/40.
Unseen programs	5 HeCBench programs the agent had never seen, 90 runs: toolkit **30/30**, no checks 18/30, a shell only 17/30.
On real GPUs	The agent's ports, unedited: **10/10** pass against the C++ reference and the original CUDA built on the same GPU, 5 on NVIDIA and 5 on an AMD Instinct MI350X.
Why checks matter	Without checks, agents shipped wrong output in 6 of 40 runs, including a softmax port that compiled cleanly and got 332,679 of 524,288 outputs wrong. With a shell, none did, but helpers stayed in 26 of 40 runs.
Cost	About the same as without: $1.37 vs $1.23 for 20 Sonnet runs.
Original examples	Claude Sonnet ported all 4 original examples to success, in 5 to 8 iterations, for $0.067 to $0.082 each. One run per example.
Hardware	Ports checked on nine NVIDIA GPU types and an AMD MI350X. On the MI350X, an in-order queue cut the softmax port's cost per launch from 114.6 to 24.4 µs (the original CUDA, converted for AMD: 8.9 µs).
Upstream	Two build bugs in Intel's open-source compiler for AMD GPUs reported as intel/llvm#23531 and #23532.

Honest limits. Only Claude models from one vendor were measured, with 3 to 5 runs per cell, on small single-file programs, graded on a CPU: indicative, not a benchmark. No port made by an outside agent over MCP has been measured; only the registration was checked. GPU kernel timings are single small launches, not a speed claim. Intel GPUs are untested.

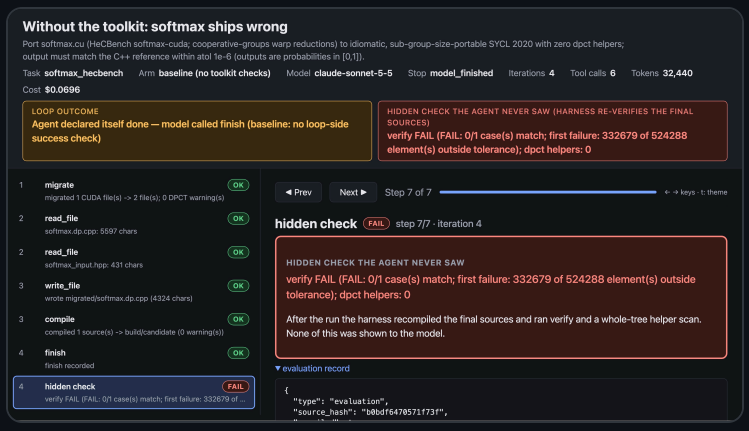
Images



1 · The README banner, redrawn for 16:9



2 · With and without the toolkit's checks



3 · Replay: a run without the toolkit, every step OK, then the hidden check fails

Supplied as 3840 × 2160 JPEGs with this kit. Image 1 is the project's README banner, redrawn. Image 2 is a new chart of the README's results. Image 3 is the project's own replay page, re-rendered at high resolution.

TO CONFIRM (Checo): no photo of the author and no video of the toolkit itself. Add a portrait if the kit needs one.

Setup and links

Install	`git clone https://github.com/checocadena/oneapi-agent-toolkit`
	`cd oneapi-agent-toolkit && pip install -e ".[all]"`
	Python 3.11 or later.

Try in a minute	`scripts/demo.sh` runs the agent loop end to end with canned tool results, marked MOCK. No toolchain, no API key.
Toolchain	For the real tools: Intel's `icpx` and SYCLomatic, installed by `scripts/install-toolchain.sh`. Without them each tool says `toolchain_missing`.
Agents	`oneapi-agent setup` registers the MCP server with Claude Code, Codex, Cursor and Gemini CLI. Other agents can use the CLI (`AGENTS.md`).
Model and GPUs	An Anthropic or Meta Model API key for the loop. A Runpod, Vast.ai, Vultr or DigitalOcean key for GPU runs.
Used in	Muse Port, by the same author: `github.com/checocadena/museport`
License	AGPL-3.0. Commercial licenses are offered for closed-source use (see the README). Versions up to commit `b08b687` were released under MIT.

Code, results and evidence: `github.com/checocadena/oneapi-agent-toolkit`
Replay demo: `checocadena.github.io/oneapi-agent-toolkit`
One-page overview: `docs/overview.md` in the repository

Contact: Checo Cadena · `github.com/checocadena` · `checocadena.com`

TO CONFIRM (Checo): the package is not on PyPI yet (the release workflow waits for a PyPI trusted publisher). Publish before sending, or keep "not yet on PyPI".

Builds on SYCLomatic (Intel), the Intel oneAPI DPC++/C++ compiler and HeCBench programs. Design inspired by REA. oneapi-agent-toolkit is an independent project and is not affiliated with Intel, NVIDIA, AMD, Anthropic or Meta.